

Imports

```
In [2]: import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd
import numpy as np
```

```
In [3]: df = pd.read_csv('C:\\Users\\saksh\\OneDrive\\Desktop\\AI-Cybersecurity-Project\\
df
```

```
Out[3]:
```

	country	year	attack_type	target_industry	financial_loss_(in_million_\$)	nun
0	China	2019	Phishing	Education	80.53	
1	China	2019	Ransomware	Retail	62.19	
2	India	2017	Man-In-The-Middle	It	38.65	
3	Uk	2024	Ransomware	Telecommunications	41.44	
4	Germany	2018	Man-In-The-Middle	It	74.41	
...	...	...	...	...	...	...
2995	Uk	2021	Ransomware	Government	51.42	
2996	Brazil	2023	Sql Injection	Telecommunications	30.28	
2997	Brazil	2017	Sql Injection	It	32.97	
2998	Uk	2022	Sql Injection	It	32.17	
2999	Germany	2021	Sql Injection	Retail	48.20	

3000 rows × 10 columns

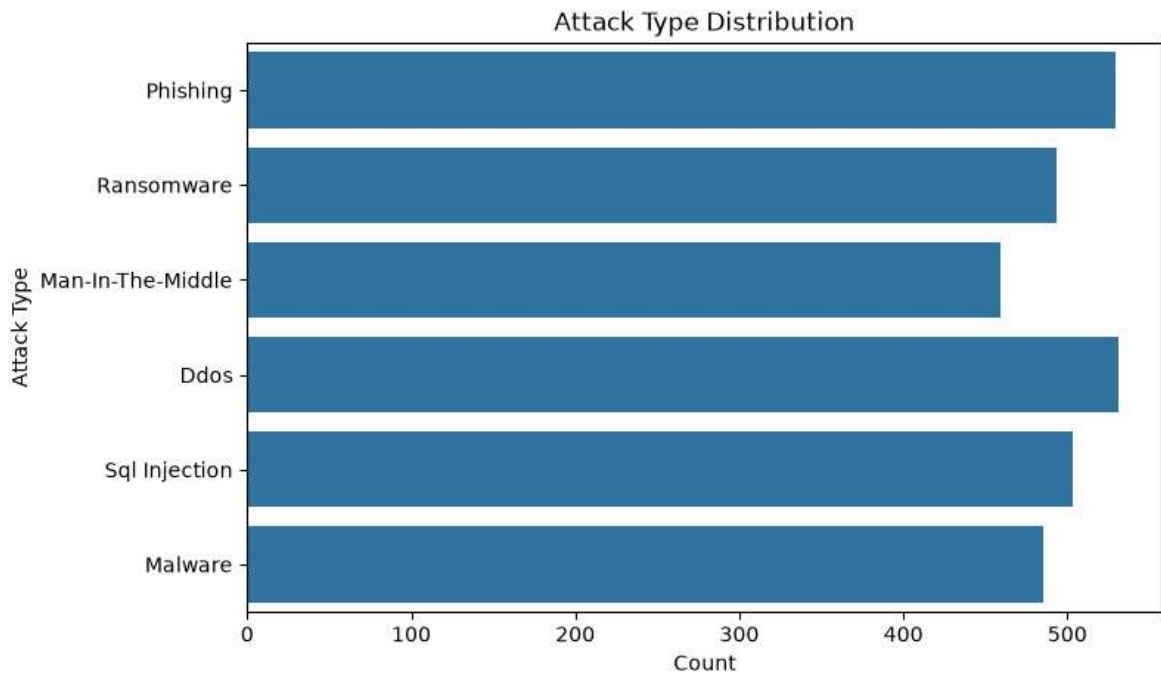


Attack type distribution

```
In [5]: df['attack_type'].value_counts()
```

```
Out[5]: attack_type
Ddos          531
Phishing      529
Sql Injection 503
Ransomware    493
Malware       485
Man-In-The-Middle 459
Name: count, dtype: int64
```

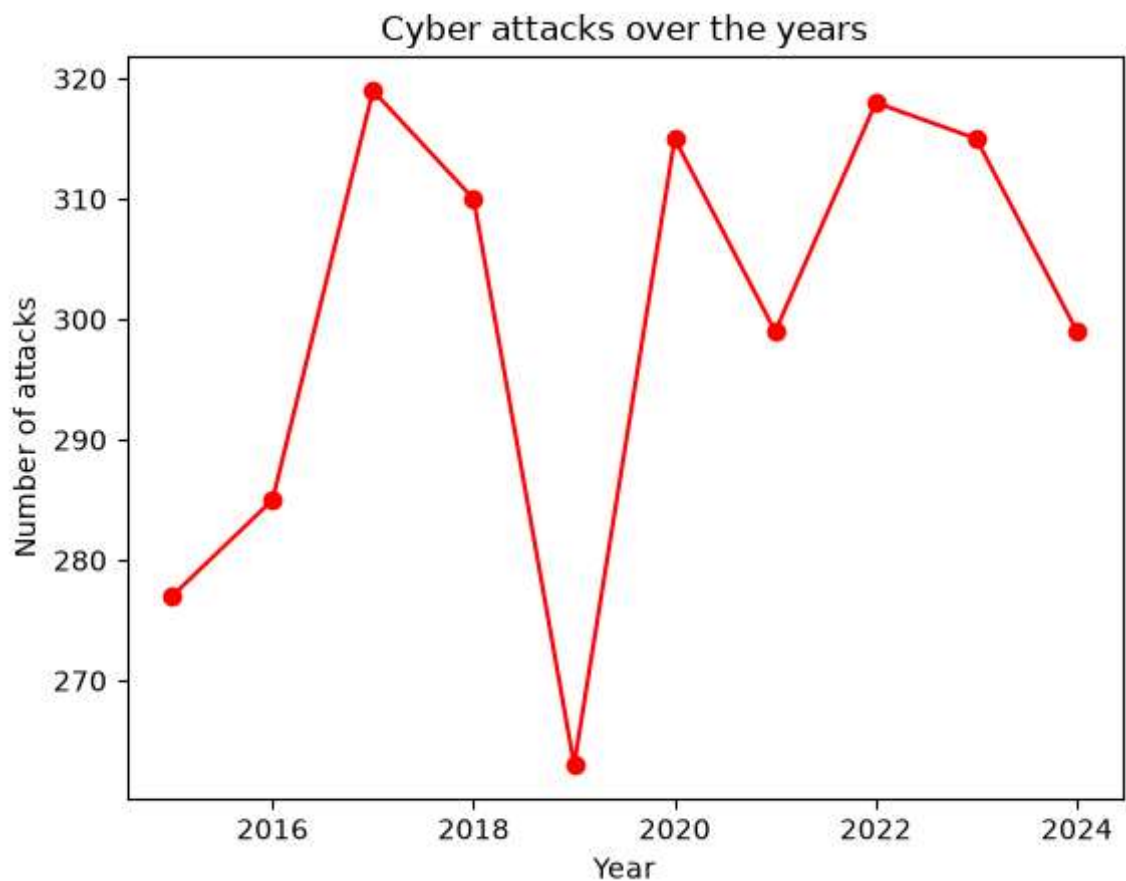
```
In [12]: plt.figure(figsize=(8,5))
sns.countplot(data=df, y='attack_type')
plt.title("Attack Type Distribution")
plt.xlabel("Count")
plt.ylabel("Attack Type")
plt.show()
```



Attacks by year

```
In [9]: yearly = df.groupby('year').size()
```

```
In [13]: yearly.plot(kind='line', color='red', marker='o')
plt.title("Cyber attacks over the years")
plt.xlabel("Year")
plt.ylabel("Number of attacks")
plt.show()
```

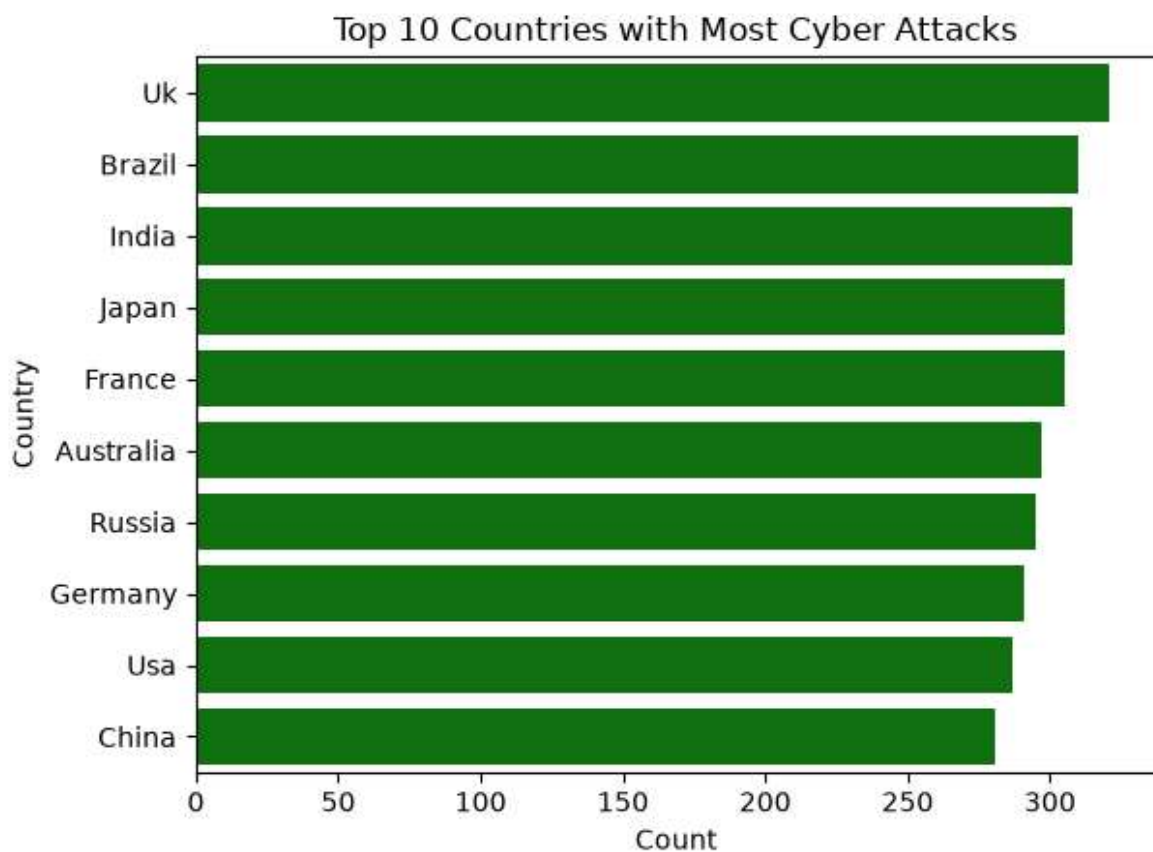


Top Countries Targeted

```
In [14]: df['country'].value_counts().head(10)
```

```
Out[14]: country
Uk          321
Brazil      310
India       308
Japan       305
France      305
Australia   297
Russia      295
Germany     291
Usa         287
China       281
Name: count, dtype: int64
```

```
In [16]: sns.countplot(data=df, y='country', color='green',
order= df['country'].value_counts().head(10).index)
plt.title("Top 10 Countries with Most Cyber Attacks")
plt.xlabel("Count")
plt.ylabel("Country")
plt.show()
```



Industry Analysis

```
In [17]: df
```

Out[17]:

	country	year	attack_type	target_industry	financial_loss_(in_million_\$)	nun
0	China	2019	Phishing	Education	80.53	
1	China	2019	Ransomware	Retail	62.19	
2	India	2017	Man-In-The-Middle	It	38.65	
3	Uk	2024	Ransomware	Telecommunications	41.44	
4	Germany	2018	Man-In-The-Middle	It	74.41	
...	...	...	...	...	...	...
2995	Uk	2021	Ransomware	Government	51.42	
2996	Brazil	2023	Sql Injection	Telecommunications	30.28	
2997	Brazil	2017	Sql Injection	It	32.97	
2998	Uk	2022	Sql Injection	It	32.17	
2999	Germany	2021	Sql Injection	Retail	48.20	

3000 rows × 10 columns

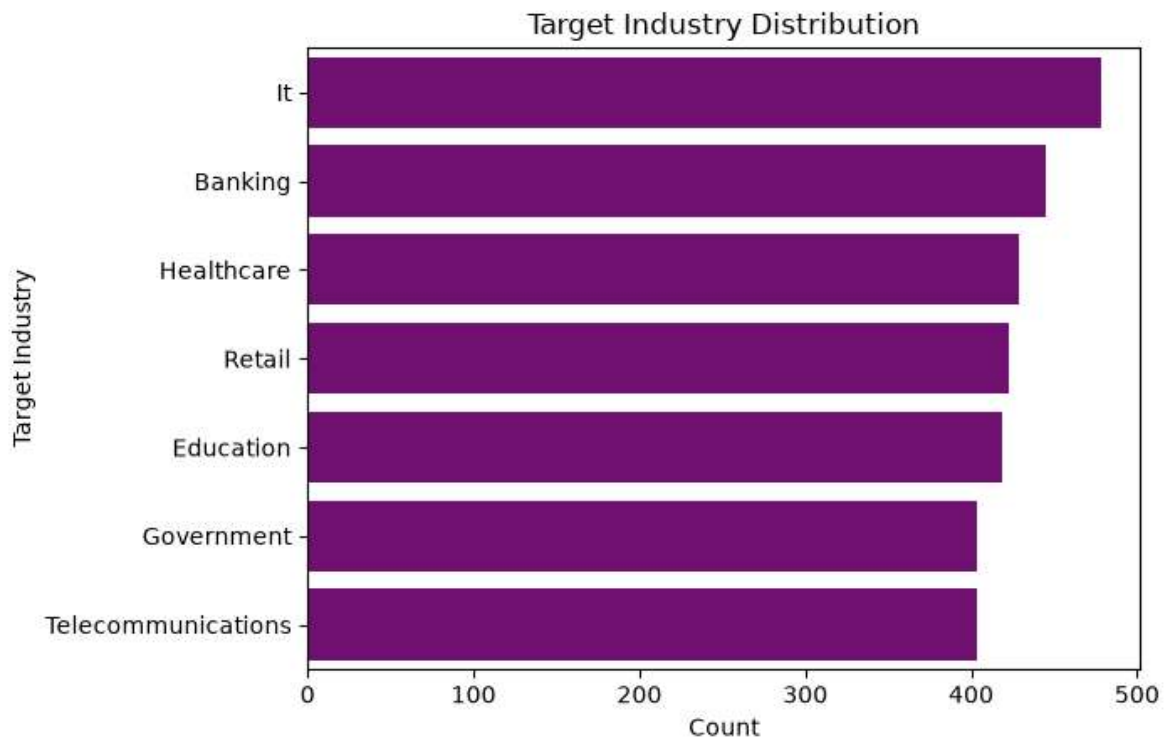
In [18]: `df['target_industry'].value_counts()`

Out[18]: target_industry

It	478
Banking	445
Healthcare	429
Retail	423
Education	419
Government	403
Telecommunications	403

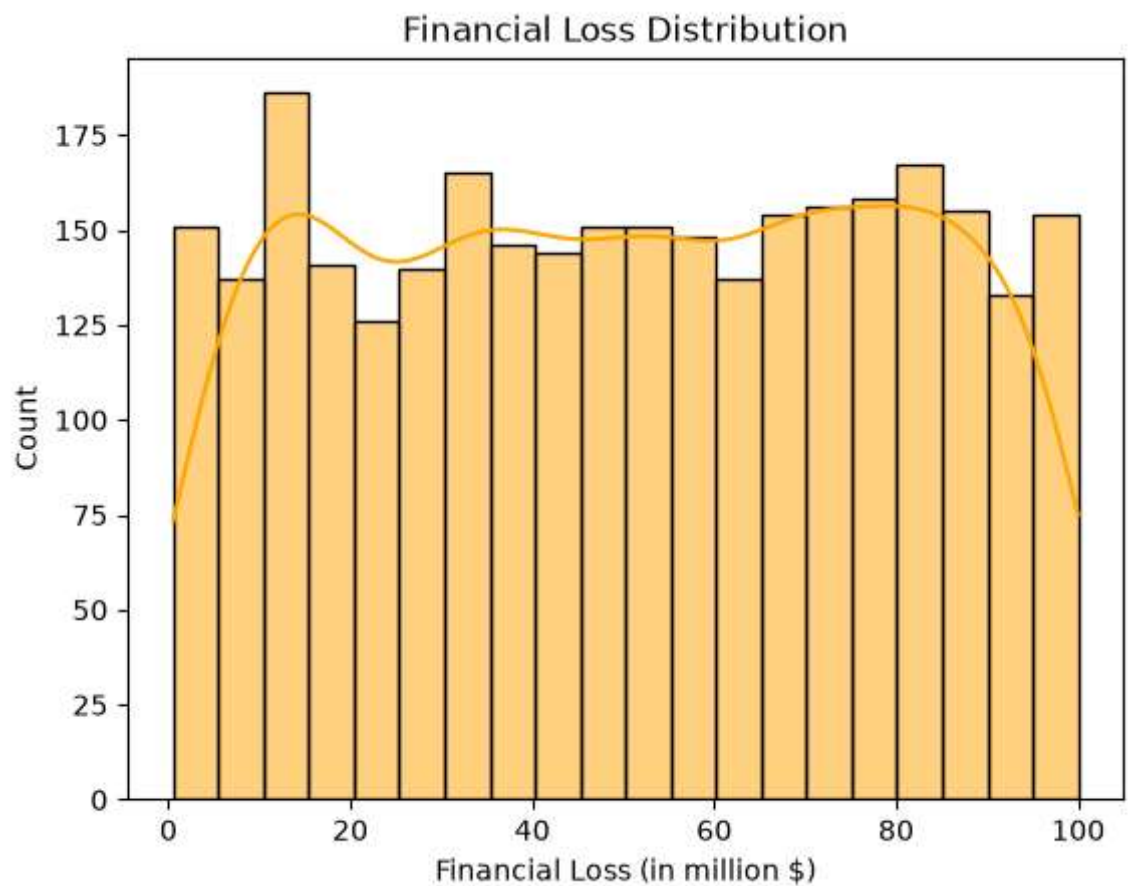
Name: count, dtype: int64

In [19]: `sns.countplot(data=df, y='target_industry', color='purple',
order=df['target_industry'].value_counts().index)
plt.title('Target Industry Distribution')
plt.xlabel('Count')
plt.ylabel('Target Industry')
plt.show()`



Financial Impact Analysis

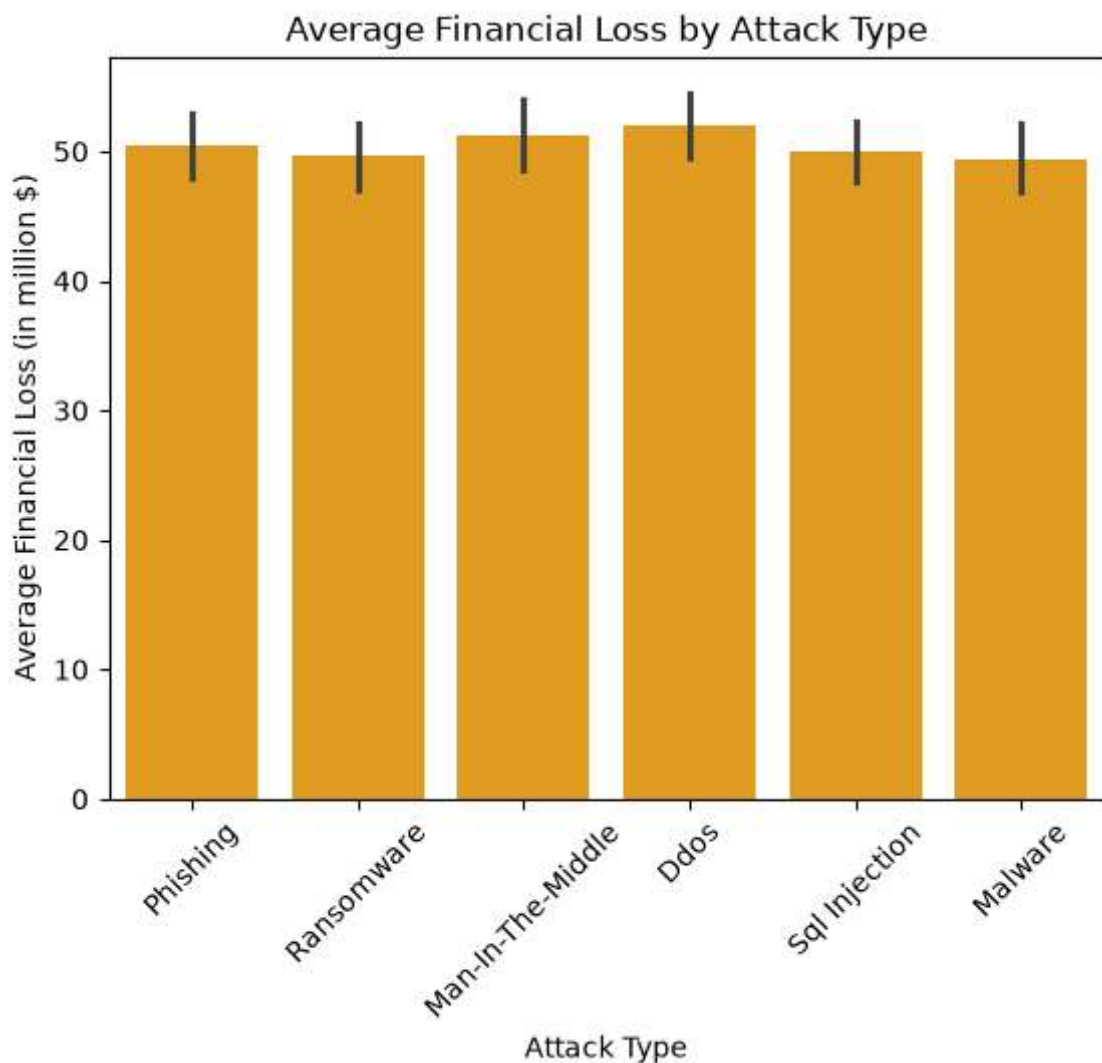
```
In [20]: sns.histplot(df['financial_loss_(in_million_$)'],  
bins = 20, color = 'orange', kde = True)  
plt.title('Financial Loss Distribution')  
plt.xlabel('Financial Loss (in million $)')  
plt.ylabel('Count')  
plt.show()
```



```
In [27]: df.groupby('attack_type')['financial_loss_(in_million_$)'].mean()
```

```
Out[27]: attack_type
Ddos                52.035631
Malware            49.418454
Man-In-The-Middle  51.309085
Phishing           50.459905
Ransomware         49.653793
Sql Injection      50.013042
Name: financial_loss_(in_million_$), dtype: float64
```

```
In [31]: sns.barplot(x='attack_type', y='financial_loss_(in_million_$)', data=df, color='
plt.title('Average Financial Loss by Attack Type')
plt.xlabel('Attack Type')
plt.ylabel('Average Financial Loss (in million $)')
plt.xticks(rotation=45)
plt.show()
```



Number of Affected Users

```
In [33]: df.groupby('attack_type')['number_of_affected_users'].mean()
```

```
Out[33]: attack_type
Ddos          499437.410546
Malware       508780.232990
Man-In-The-Middle 520064.320261
Phishing      487179.536862
Ransomware    502825.369168
Sql Injection 512469.833002
Name: number_of_affected_users, dtype: float64
```

Top Incidents by affected users

```
In [34]: df.nlargest(10, 'number_of_affected_users')
```

```
Out[34]:
```

	country	year	attack_type	target_industry	financial_loss_(in_million_\$)	num
2045	Usa	2023	Ransomware	Government	93.34	
323	France	2022	Sql Injection	Healthcare	27.06	
2240	India	2020	Malware	It	84.43	
2004	Usa	2016	Ransomware	Telecommunications	99.88	
2440	Usa	2020	Ddos	Banking	60.20	
1515	India	2024	Ransomware	Government	79.30	
2852	China	2021	Ransomware	Retail	40.01	
2775	Uk	2023	Man-In-The-Middle	Banking	24.28	
928	France	2019	Phishing	Retail	25.75	
1561	Usa	2016	Phishing	Retail	89.85	

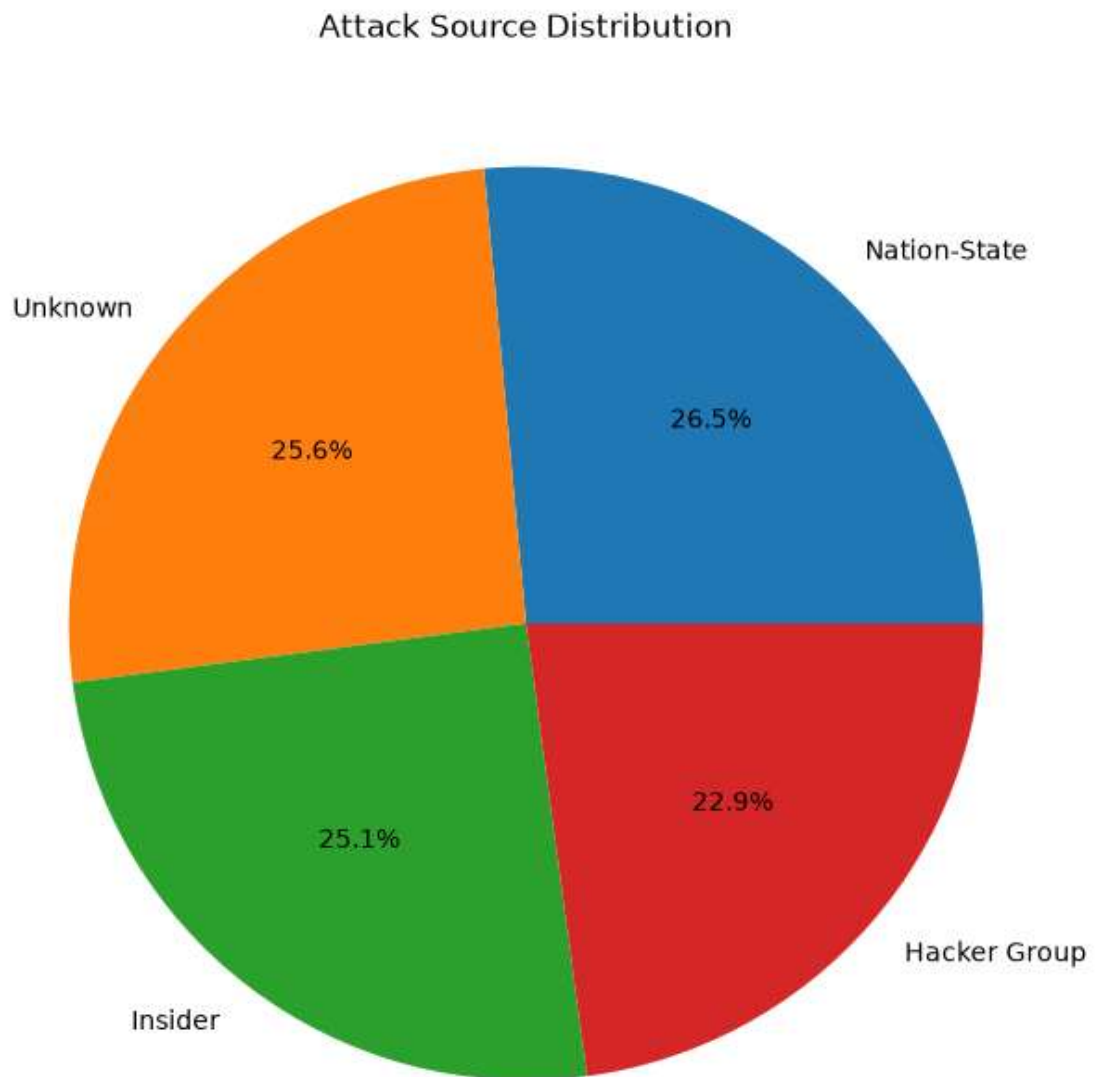


Attack Source Analysis

```
In [59]: attack_source_counts = df['attack_source'].value_counts()
attack_source_counts
```

```
Out[59]: attack_source
Nation-State    794
Unknown         768
Insider         752
Hacker Group    686
Name: count, dtype: int64
```

```
In [60]: plt.figure(figsize=(8,8))
plt.pie(
    attack_source_counts,
    labels=attack_source_counts.index,
    autopct='%1.1f%%',
)
plt.title('Attack Source Distribution')
plt.show()
```



Security Vulnerability Analysis

```
In [39]: df['security_vulnerability_type'].value_counts()
```

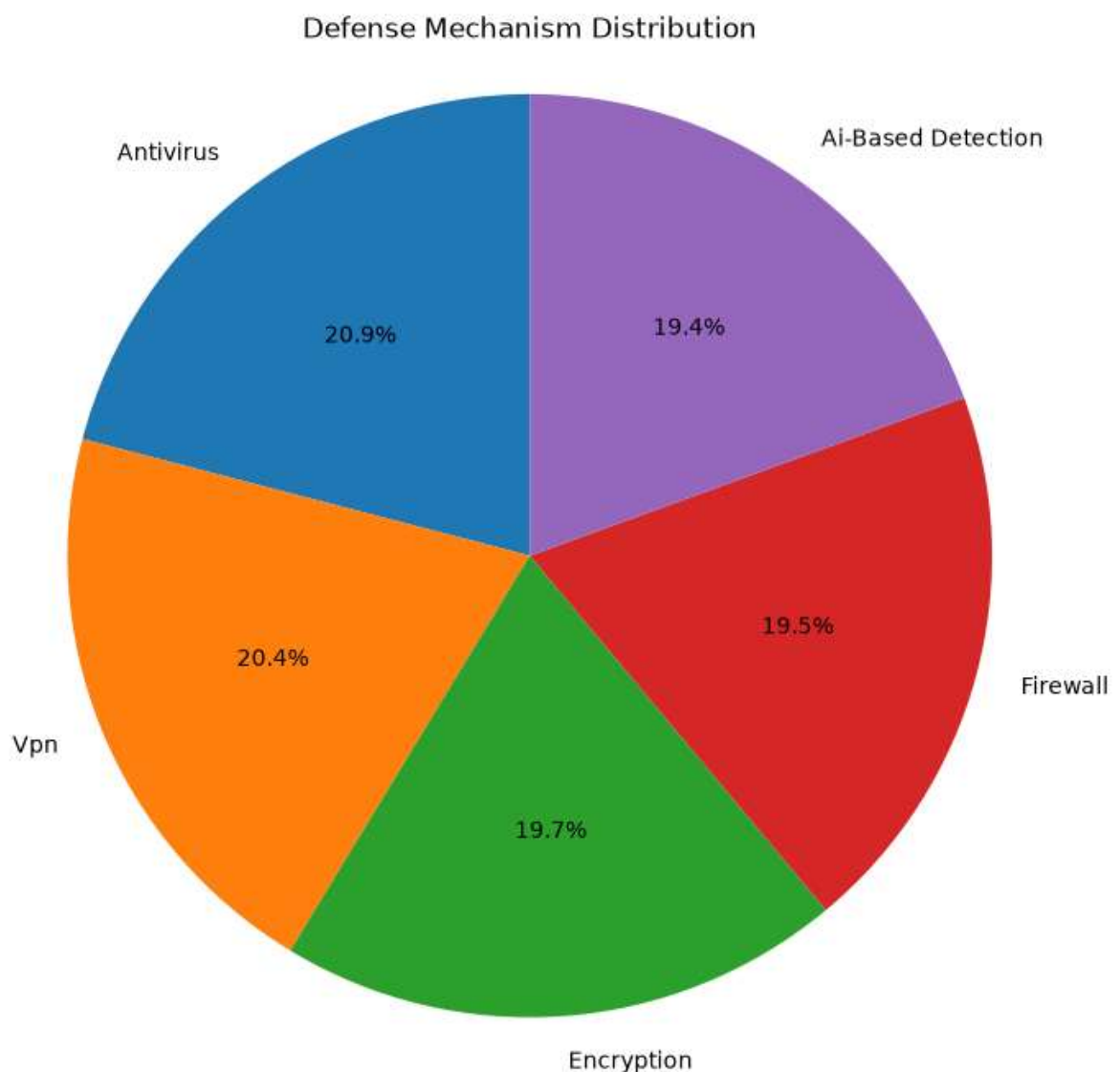
```
Out[39]: security_vulnerability_type
Zero-Day      785
Social Engineering  747
Unpatched Software  738
Weak Passwords  730
Name: count, dtype: int64
```

Defence mechanism analysis

```
In [61]: defense_counts = df['defense_mechanism_used'].value_counts()
defense_counts
```

```
Out[61]: defense_mechanism_used
Antivirus      628
Vpn            612
Encryption     592
Firewall       585
Ai-Based Detection  583
Name: count, dtype: int64
```

```
In [64]: plt.figure(figsize=(8,8))
plt.pie(
    defense_counts,
    labels=defense_counts.index,
    autopct='%1.1f%%',
    startangle=90
)
plt.title('Defense Mechanism Distribution')
plt.axis('equal') # Equal aspect ratio ensures that pie is drawn as a circle.
plt.show()
```



Defence Mechanism vs financial loss

```
In [41]: df.groupby('defense_mechanism_used')['financial_loss_(in_million_$)'].mean()
```

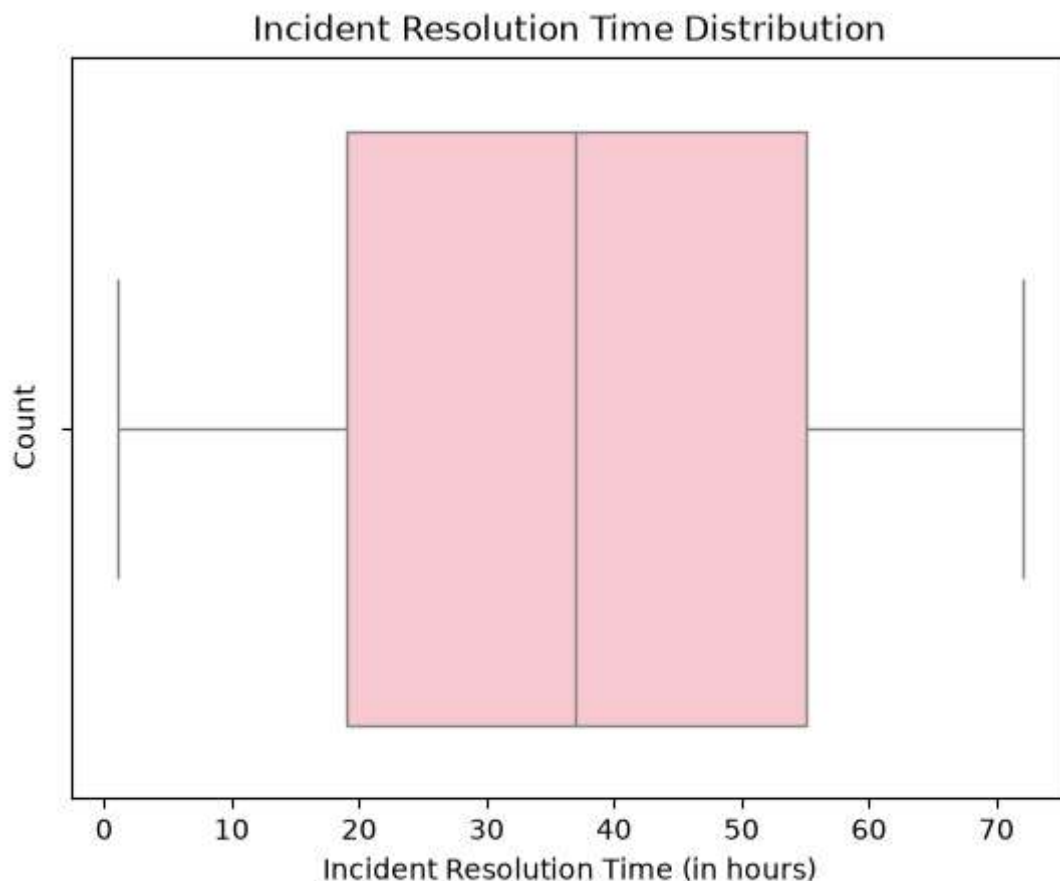
```
Out[41]: defense_mechanism_used
Ai-Based Detection    50.361527
Antivirus             51.698838
Encryption            50.400203
Firewall              49.719709
Vpn                   50.209673
Name: financial_loss_(in_million_$), dtype: float64
```

Response time analysis

```
In [45]: df['incident_resolution_time_(in_hours)'].mean()
```

```
Out[45]: np.float64(36.476)
```

```
In [21]: sns.boxplot(
        x=df['incident_resolution_time_(in_hours)'], color='pink'
    )
plt.title('Incident Resolution Time Distribution')
plt.xlabel('Incident Resolution Time (in hours)')
plt.ylabel('Count')
plt.show()
```



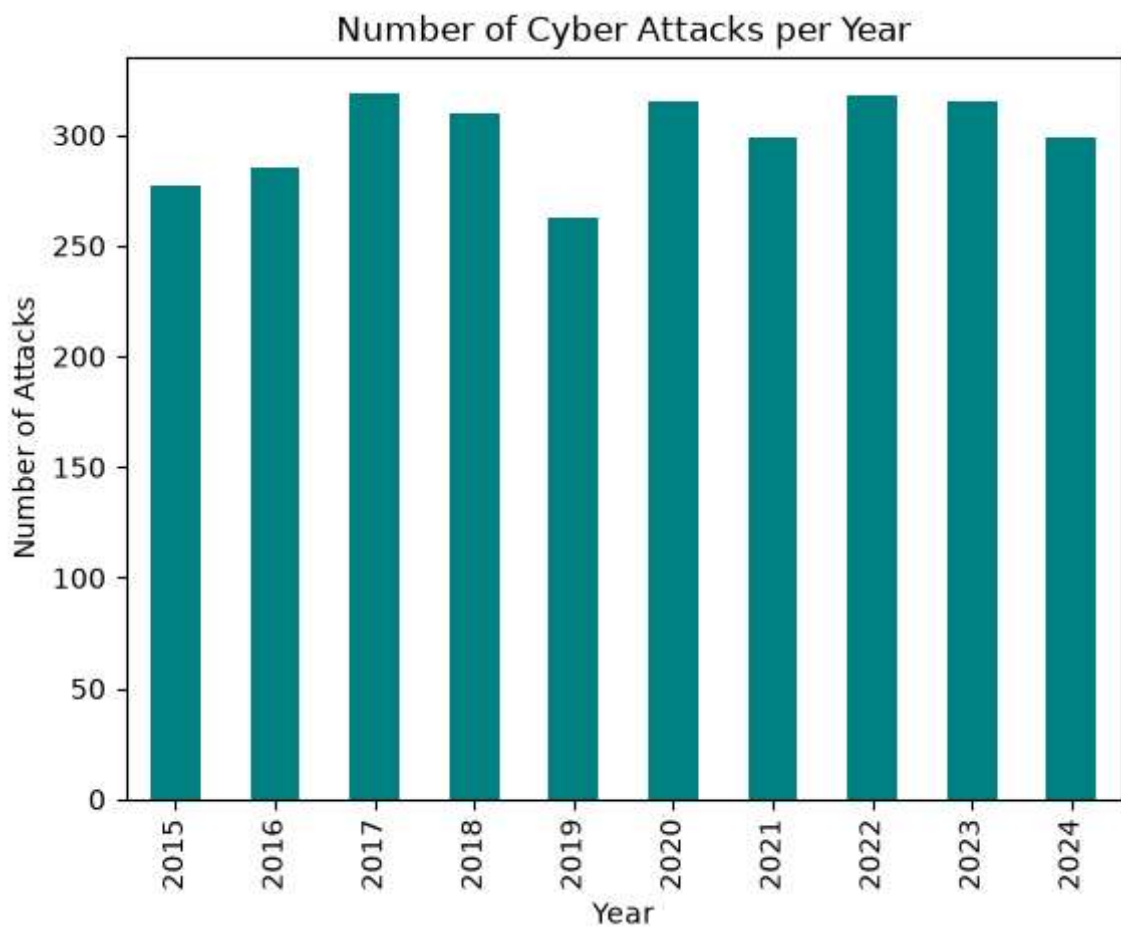
Resolution time by attack type

```
In [46]: df.groupby('attack_type')['incident_resolution_time_(in_hours)'].mean()
```

```
Out[46]: attack_type
Ddos      35.687382
Malware   37.074227
Man-In-The-Middle  36.871460
Phishing  35.913043
Ransomware  36.533469
Sql Injection  36.906561
Name: incident_resolution_time_(in_hours), dtype: float64
```

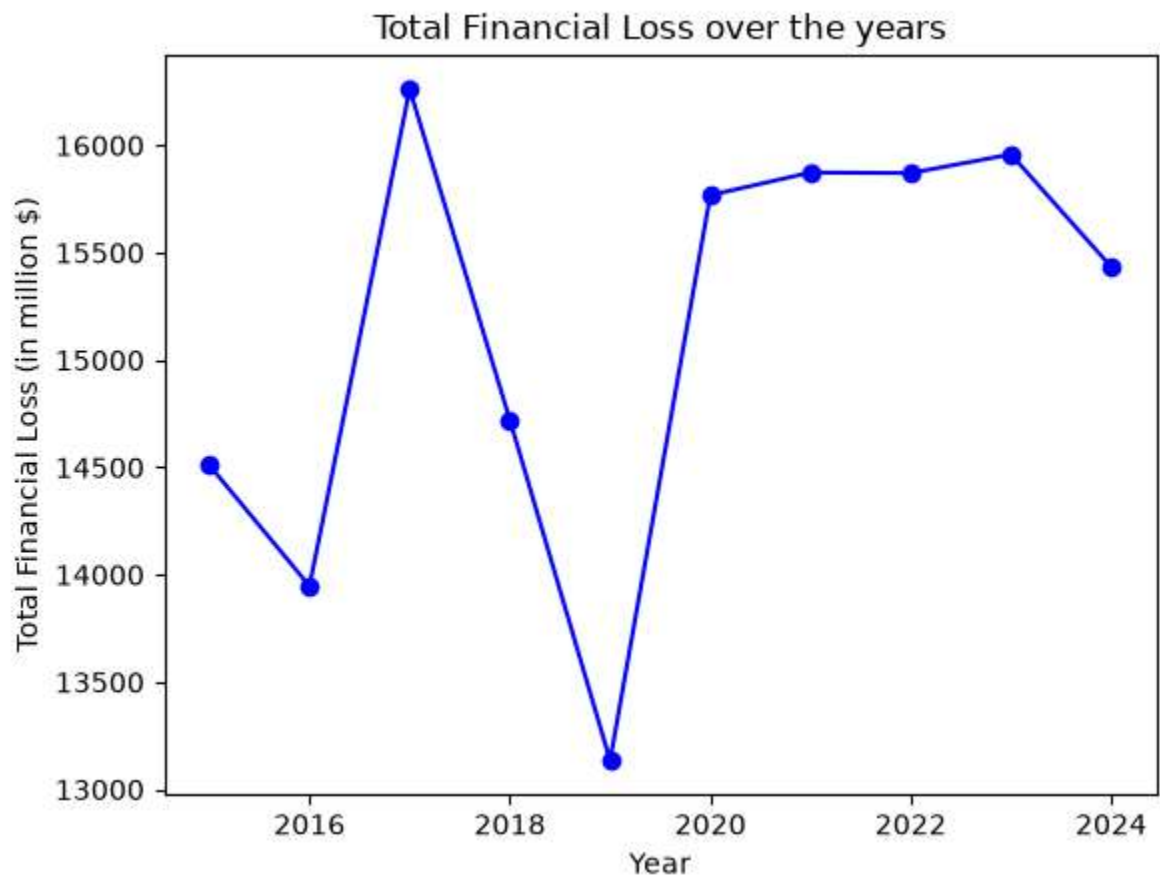
Trend Analysis by year

```
In [50]: df.groupby('year').size().plot(kind='bar', color='teal')
plt.title('Number of Cyber Attacks per Year')
plt.xlabel('Year')
plt.ylabel('Number of Attacks')
plt.show()
```



Financial Loss trend

```
In [49]: df.groupby('year')['financial_loss_(in_million_$)'].sum().plot(kind='line', color='red')
plt.title("Total Financial Loss over the years")
plt.xlabel("Year")
plt.ylabel("Total Financial Loss (in million $)")
plt.show()
```



Correlation Analysis

```
In [51]: numeric_cols = [
    'financial_loss_(in_million_$)',
    'number_of_affected_users',
    'incident_resolution_time_(in_hours)'
]
```

```
In [ ]: df[numeric_cols].corr()
```

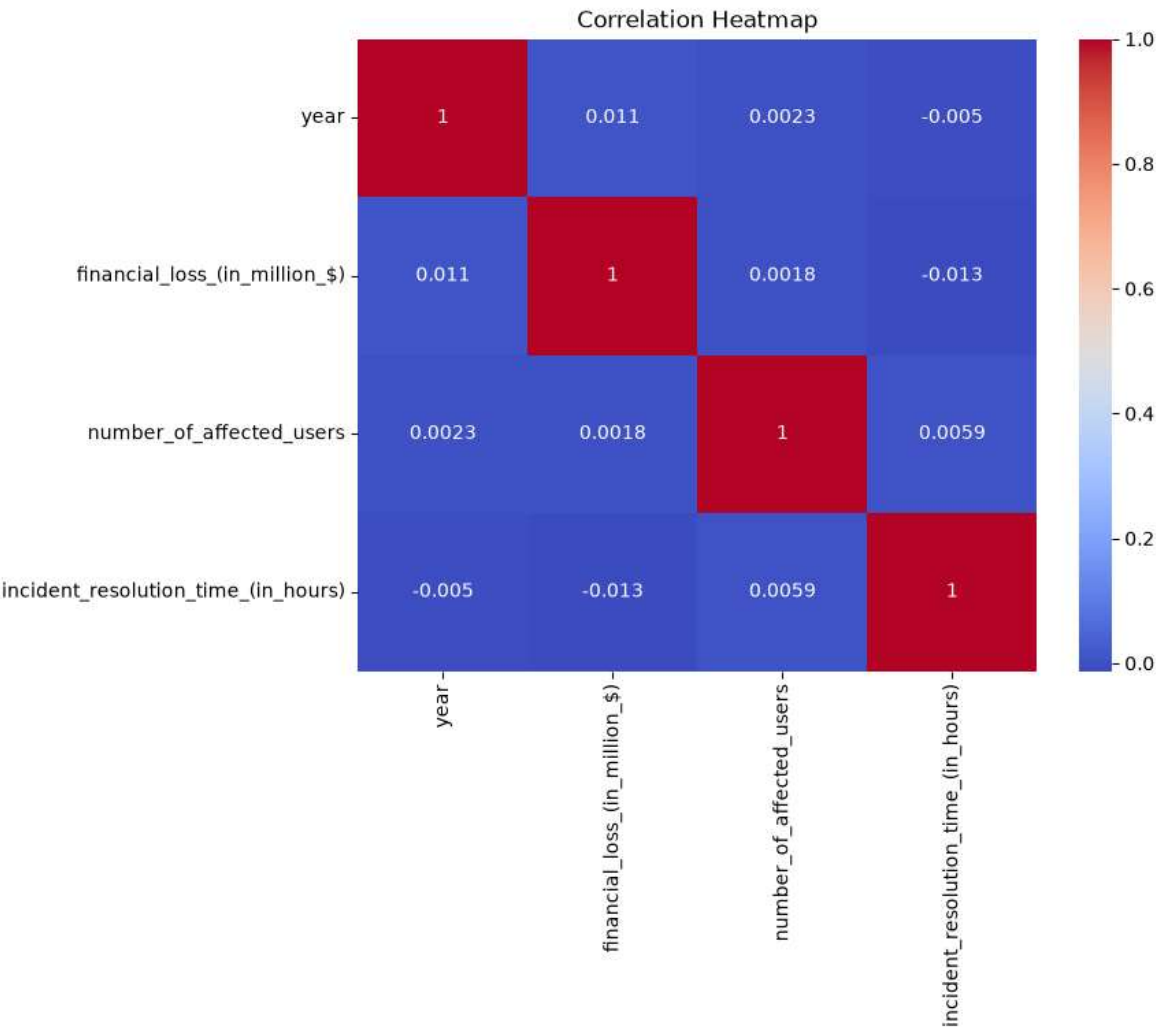
```
Out[ ]:
```

	financial_loss_(in_million_\$)	number_of_affected_users
financial_loss_(in_million_\$)	1.000000	0.001787
number_of_affected_users	0.001787	1.000000
incident_resolution_time_(in_hours)	-0.012671	0.005893

```
In [22]: numeric_df = df.select_dtypes(include=['int64', 'float64'])
```

```
In [53]: corr = numeric_df.corr()
```

```
In [54]: plt.figure(figsize=(8,6))
sns.heatmap(
    corr,
    annot=True,
    cmap='coolwarm')
plt.title('Correlation Heatmap')
plt.show()
```



In []: